

POWERED BY **ZAIZENS** · AN INTEGRATED HMS

PREMIUM · PROFESSIONAL EDITION

Pharmacy & Dispensing

Complete staff guide — validate PHA codes, dispense medications, inventory, off-catalog pricing, nursing requests, purchase orders, intelligence reports, and cashier integration.

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Pharmacy User Manual

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Introduction

Welcome to the **ZAIZENS Integrated HMS Pharmacy User Manual**. The Pharmacy module manages the full medication pathway — from doctor prescriptions and cashier payment through validation, stock checks, dispensing, and inventory intelligence.

Pharmacy hub: One workspace at `/pharmacy` with Dashboard, Products, Dispensing, and Prescriptions tabs — plus validate-and-dispense at `/pharmacy/validate`.

PHA
SERVICE CODES

4
HUB TABS

8+
INTELLIGENCE REPORTS



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1. Pharmacy Overview & Roles

The **Pharmacy module** connects clinical prescriptions to verified, billable, and dispensed medications across OPD, IPD, Emergency, and ward supply contexts.

**Pharmacist**
Validate PHA codes, check stock, dispense line-by-line, counsel patients

**Doctor**
Prescribes medications from consultation, IPD, or ER chart

**Cashier**
Collects payment; issues PHA service codes on order slips

**Inventory / Procurement**
Stock receive, purchase orders, reorder levels, vendor management

**Nurse**
Ward supply requests; receives dispensed items from pharmacy

**Patient**
Pays at cashier, presents PHA code at pharmacy window, collects medicines

Key URLs

Screen	Path
Pharmacy Portal	/portal/pharmacy
Pharmacy Hub	/pharmacy
Validate & dispense	/pharmacy/validate
Prescription Registry	/prescriptions
Order alerts	/pharmacy/order-alerts
Nursing requests	/pharmacy/nursing-requests
Purchase orders	/pharmacy/purchase-orders
Intelligence reports	/pharmacy/reporting
Expiry report	/pharmacy/reporting/expiry
Inventory (central)	/inventory
Service catalog	/catalog

Permissions: `pharmacy.read` view stock & queue · `pharmacy.write` dispense & adjust stock · `prescription.read/write` prescription registry

2. End-to-End Dispensing Workflow

Doctor prescribes → Patient pays at Cashier (PHA code issued)
→ Pharmacy validates code → Stock check → Dispense line-by-line
→ Patient receives medicines → Chart updated



Standard OPD pathway

- 1 Doctor adds medication(s) during consultation — dosage, frequency, duration.
- 2 Patient pays at Cashier — receives slip with **PHA-...** service code.
- 3 Patient presents PHA code at pharmacy window.
- 4 Pharmacist validates code — reviews each medication line.
- 5 Check stock on hand; warn if expired or insufficient quantity.
- 6 Dispense one line at a time — stock deducted automatically.
- 7 Partial Rx: remaining lines stay pending until next visit.

Off-catalog exception

Custom prescriptions with no catalog price require pharmacist to mark available and set unit price before patient pays at Cashier.

Payment gate: In-house medications cannot be dispensed until the order line is paid (unless local policy allows credit for IPD/ER).

3. Pharmacy Portal

The **Pharmacy Portal** (`/portal/pharmacy`) is the pharmacist home screen.

KPI strip

- **Total Prescriptions** — all registered prescriptions
- **Issued Today** — new prescriptions today
- **Active / Pending** — awaiting dispense
- **Dispensed** — completed dispensing
- **Nursing requests** — ward supply requests pending

Quick actions

- **Prescriptions Queue** — `/prescriptions`
- **Order alerts** — new doctor prescriptions
- **Nursing requests** — ward supply with badge count when pending

Portal tiles link to pharmacy hub, validate, reporting, and configuration shortcuts.

4. Pharmacy Hub Tabs

The revamped **Pharmacy Hub** at `/pharmacy` organizes daily work in four tabs:



Dashboard

KPIs: products count, awaiting dispense, dispensed today, out of stock, active Rx, Rx today. Quick links to products, dispensing queue, validate, and expiry report.

Products

Pharmacy inventory linked to service catalog — SKU, category, price, on-hand quantity, reorder level. Stock adjust (+/-), receive stock, sync from catalog.

Dispensing

Toggle **Dispensed log** (by day) vs **Awaiting** pending lines. Search by patient, medication, PHA code, pharmacist.

Prescriptions

Historical prescription registry — Rx ID, patient, title, status, created date.

5. Validate PHA Code & Dispense

The validate screen is the primary dispensing entry point when a patient presents a payment slip.

- 1 Open `/pharmacy/validate` or click **Validate & dispense** from hub.
- 2 Enter **PHA service code** (e.g. `PHA-7721-M3KP8X2Q`).
- 3 System verifies payment and loads all medication lines on the code.
- 4 Review patient, referring doctor, and prescription status badge.
- 5 For each in-house line: check dosage, quantity, stock, expiry.
- 6 Click **Dispense this medication** per line — one at a time.
- 7 Add pharmacist notes (batch, substitution, counselling) if needed.

Line status badges

- NOT YET DISPENSED
- PAID · AWAITING
- DISPENSED
- OFF-CATALOG

Prescription progress

Header shows **Partially dispensed (2 of 5)** or **Fully dispensed** as lines are completed.

Stock warnings

- Insufficient stock — option to dispense anyway (stock may go negative for off-catalog)
- Expired batch — dispense blocked or warning per policy
- No inventory SKU linked — off-catalog or catalog sync needed

6. Order Alerts

Order alerts (`/pharmacy/order-alerts`) surface doctor-prescribed medications from consultations, IPD, and Emergency.

- 1

Open Order alerts from portal or sidebar.
- 2

Review unacknowledged prescriptions needing pharmacy attention.
- 3

Each alert shows patient, medication, context (OPD / IPD / Emergency).
- 4

Mark as seen when handled via validate flow or off-catalog pricing.

Clinical department alerts strip on portal highlights pharmacy items requiring action.

7. Off-Catalog & Custom Prescriptions

When a doctor prescribes a medication **not in the hospital catalog** or with zero price:

- 1 Line appears on validate screen with **Custom Rx** or **Off-catalog** badge.
- 2 Pharmacist toggles **Available off-catalog** if product can be supplied.
- 3 Enter **unit price (FCFA)** and save — patient sent to Cashier on same PHA code.
- 4 After payment, re-validate same PHA code to dispense.
- 5 Stock may go negative until inventory is updated for off-catalog items.

Cashier billing panel

Cashier OPD bill shows **Await pharmacy** badge until pharmacist confirms off-catalog availability and price.

Clear off-catalog toggle if product cannot be supplied — patient may need refund or alternative prescription.

8. External Medications

Medications the patient obtains **outside the hospital** are documented without dispensing:

- 1 Doctor marks line as external on prescription.
- 2 Validate screen shows external section separate from in-house items.
- 3 Pharmacist clicks **Acknowledge served outside** — no stock movement.
- 4 Chart shows line resolved with "served outside hospital" note.

No payment collection at cashier for external-only lines unless local policy requires documentation fee.

9. Products & Inventory

Product list

Products tab shows pharmacy catalog items with inventory linkage:

Column	Description
SKU	Stock keeping unit code
Product	Medication name
Category	Medicine type / therapeutic category
Price	Tariff from Service Catalog
On hand	Current stock quantity
Reorder at	Minimum level before alert

Stock operations

- **Adjust** — +/- 1 or custom delta
- **Receive** — add quantity from goods receipt
- **Reorder level** — set par level for low-stock alerts
- **Sync from catalog** — import new catalog items to inventory
- **Stock history** — view movements per product

Central inventory

`/inventory` provides hospital-wide SKU management — import pharmacy catalog, apply stock sheet, RFQ for restock.

Stock status

IN STOCK · **LOW STOCK** · Out of stock (critical alert on dashboard)

10. Dispensing Queue & Log

Awaiting tab

All paid prescription lines not yet dispensed — open validate via PHA code link.

Dispensed log

Filter by date — shows patient, medication, PHA code, quantity, pharmacist, dispensed timestamp.

Legacy IPD path

Older IPD prescription lines may appear in legacy queue — dispense via line ID action when applicable.

Dispense with expiry warning

If batch is near expiry, system may redirect through `/pharmacy/dispense-warn/:lineId` — confirm before proceeding.

11. Prescription Registry

Prescription Registry at `/prescriptions` lists all clinical medication orders.

Column	Description
Rx ID	Prescription reference number
Patient	Name and ID
Order title	Prescription summary
Status	Active, Partial, Dispensed, Completed, Cancelled
Created	Order date

Actions

- **View Details** — full prescription with all lines
- **Print Rx** — patient copy for external pharmacy if needed
- **Patient Chart** — navigate to clinical record

12. Nursing Supply Requests

Ward nurses request supplies via **Nursing requests** (</pharmacy/nursing-requests>).

1 Nurse submits supply request from ward — items and quantities needed.

2 Pharmacy portal shows badge count when requests pending.

3 Pharmacist reviews request — mark preparing or dispense.

4 Stock deducted on dispense; nurse acknowledges receipt on ward.

Integrates with IPD ward board and nursing module supply workflow.

13. Purchase Orders

Pharmacy procurement at </pharmacy/purchase-orders> :

- 1

New PO — select vendor, add product lines with quantities and unit costs.
- 2

Draft — review before confirmation.
- 3

Confirm — PO sent to vendor; cannot edit lines freely.
- 4

Receive goods — record delivery; stock updated automatically.

Links to procurement vendors and inventory movement audit trail.

Purchase activity report available under Intelligence Reports.

14. Intelligence Reports

Pharmacy Intelligence Reports hub at </pharmacy/reporting> — premium analytics with period filters and print.

Report	Purpose
Stock movement	In/out movements over period
Dispensing	Medications dispensed by day/pharmacist
Fast-moving	Top sellers for reorder planning
Dormant stock	Slow-moving items tying up capital
Low stock	Items at or below reorder level
Stock valuation	Inventory value at current prices
Purchase activity	PO and receiving summary
Negative stock	SKUs below zero — data quality alert
Expiry	Batches expiring within N days

Each report supports period selection (today, week, month, custom) and print-friendly layout.

15. Expiry Management

Prevent dispensing expired medications:

- **Expiry report** — `/pharmacy/reporting/expiry` lists batches nearing expiry
- Validate screen shows expiry hint per order line when batch data linked
- Dispense blocked or warned for expired stock
- Remove expired stock via inventory adjustment after physical quarantine

Configure expiry window (days) on expiry report filter.

16. Cashier & Wallet Integration

Pharmacy depends on Cashier for payment verification:

- 1 Doctor order creates pharmacy order line — pending payment.
- 2 Cashier collects fee; marks line paid; PHA code on receipt.
- 3 Pharmacist validates — system confirms paid status before dispense.
- 4 **Wallet** — patient may pay via pre-paid wallet; same PHA validate flow.
- 5 Partial payment — patient returns to cashier for remaining lines.

Off-catalog items: pharmacist sets price → cashier bills → patient pays → return to pharmacy.

See **Cashier User Manual** (</docs/cashier-users-manual>) for payment methods and code validity.

17. Catalog & Configuration

Service catalog

Pharmacy medications and XAF tariffs at `/catalog` → Pharmacy department. Import 139-item price list or Excel/PDF/Word file.

Medicine types & categories

- `/pharmacy/config/medicine-types` — tablet, syrup, injection, etc.
- `/pharmacy/config/medicine-categories` — therapeutic class; Rx-required flag

Product profiles

Link inventory SKU to medicine type, category, and catalog item on product profile save.

Sync inventory

Hub → Products → **Sync products from catalog** — adds new catalog items; keeps existing quantities.

18. Screen Reference (UI)

●●● Pharmacy Portal — /portal/pharmacy

Pharmacy Portal

Validate · Dispense · Manage stock · Reports

Rx 156

Pending 9

Dispensed 142

Nursing 2

Prescriptions

Order alerts

Nursing requests

●●● Validate & dispense — /pharmacy/validate

Pharmacy · Validate code

PHA-7721-M3KP8X2Q

Amoxicillin 500mg · Qty 21 · Stock 450

Dispense this medication

Paracetamol 500mg · ✓ Dispensed

Rx status: Partially dispensed (1 of 2)

●●● Pharmacy Hub — /pharmacy

Dashboard

Products

Dispensing

Prescriptions

139

Products

9

Awaiting

24

Today

3

Out

12

Active Rx

Live screenshots: Open /portal/pharmacy or /pharmacy while signed in; Print → Save as PDF for training.

19. Troubleshooting

Issue	Solution
Code rejected — payment required	Patient must pay at Cashier; verify order line paid status
Wrong department code	PHA for pharmacy; LAB and RAD go to respective departments
Cannot dispense — awaiting pharmacist price	Set off-catalog price first; send patient to Cashier
Insufficient stock	Receive stock or use force dispense for off-catalog; create PO
Expired medication blocked	Check expiry report; quarantine batch; dispense from valid batch
Product not in inventory	Sync from catalog or add SKU in inventory module
Partial Rx stuck	Re-validate same PHA code; dispense remaining lines
Cashier shows "Await pharmacy"	Confirm off-catalog availability and enter unit price
Nursing request missing	Verify nurse submitted from ward; check nursing-requests page
Negative stock alert	Run negative stock report; receive goods or adjust data

20. Glossary

Term	Definition
PHA code	Service code prefix for pharmacy order lines (e.g. PHA-7721-...)
Validate	Verify payment and open prescription for dispensing
Off-catalog	Medication available but not in standard inventory catalog
Custom Rx	Doctor-prescribed item with no preset catalog price
External medication	Patient obtains outside hospital — acknowledge only
SKU	Stock keeping unit — inventory identifier
Reorder level	Minimum stock before low-stock alert
Partial dispense	Some lines dispensed; others remain pending
PO	Purchase order — procurement document for restocking
Order alert	Notification of new doctor prescription
Nursing request	Ward supply requisition fulfilled by pharmacy